

► CLASSIC PRESENTATIONS

CLINICAL PRESENTATION	DIAGNOSIS/DISEASE	PAGE
Gout, intellectual disability, self-mutilating behavior in a boy	Lesch-Nyhan syndrome (HGPRT deficiency, X-linked recessive)	35
Situs inversus, chronic ear infections, sinusitis, bronchiectasis, infertility	Primary ciliary dyskinesia (Kartagener syndrome)	47
Blue sclera, multiple fractures, dental problems, conductive/mixed hearing loss	Osteogenesis imperfecta (type I collagen defect)	49
Elastic skin, joint hypermobility, bleeding tendency	Ehlers-Danlos syndrome (type V collagen defect, type III collagen defect seen in vascular subtype)	49
Arachnodactyly, lens dislocation (upward and temporal), aortic dissection, hyperflexible joints	Marfan syndrome (defective fibrillin-1)	50
Arachnodactyly, pectus deformity, lens dislocation (downward and nasal)	Homocystinuria (autosomal recessive cystathionine synthase deficiency)	50
Café-au-lait spots (unilateral), polyostotic fibrous dysplasia, precocious puberty, multiple endocrine abnormalities	McCune-Albright syndrome (G_s -protein activating mutation)	55
Meconium ileus in neonate, recurrent pulmonary infections, nasal polyps, pancreatic insufficiency, infertility/subfertility, malabsorption/vitamin deficiencies	Cystic fibrosis (<i>CFTR</i> gene defect, chromosome 7, $\Delta F508$)	58
Calf pseudohypertrophy	Muscular dystrophy (most commonly Duchenne, due to X-linked recessive frameshift mutation of dystrophin gene)	59
Child uses arms to stand up from squat	Duchenne muscular dystrophy (Gowers sign)	59
Slow, progressive muscle weakness in boys	Becker muscular dystrophy (X-linked non-frameshift deletions in dystrophin; less severe than Duchenne)	59
Infant with cleft lip/palate, microcephaly or holoprosencephaly, polydactyly, cutis aplasia	Patau syndrome (trisomy 13)	61
Infant with microcephaly, rocker-bottom feet, clenched hands, structural heart defect	Edwards syndrome (trisomy 18)	61
Single palmar crease, flat facies, prominent epicanthal folds, congenital heart disease, intellectual disability	Down syndrome (trisomy 21)	61
Microcephaly, high-pitched cry, intellectual disability	Cri-du-chat (cry of the cat) syndrome; congenital deletion on short arm of chromosome 5	62
Confusion, ophthalmoplegia/nystagmus, ataxia	Wernicke encephalopathy (add confabulation/memory loss and personality changes for Korsakoff syndrome)	64
Dilated cardiomyopathy/high-output heart failure, edema, alcoholism or malnutrition	Wet beriberi (thiamine [vitamin B_1] deficiency)	64
Dermatitis, dementia, diarrhea	Pellagra (niacin [vitamin B_3] deficiency)	65
“Burning feet syndrome,” dermatitis, enteritis, alopecia	Pantothenic acid (vitamin B_5) deficiency	65
Megaloblastic anemia, subacute combined degeneration, paresthesias, cognitive changes	Cobalamin (vitamin B_{12}) deficiency; malabsorption, decreased intrinsic factor, absent terminal ileum	67
Swollen gums, mucosal bleeding, poor wound healing, petechiae, corkscrew hairs, perifollicular hemorrhages	Scurvy (vitamin C deficiency: inability to hydroxylate proline/lysine for collagen synthesis); tea and toast diet	67

CLINICAL PRESENTATION	DIAGNOSIS/DISEASE	PAGE
Bowlegs (children), bone pain, and muscle weakness	Vitamin D deficiency: rickets (children), osteomalacia (adults); ↓ sun exposure, chronic kidney disease	68
Hemorrhagic disease of newborn with ↑ aPTT, normal bleeding time	Vitamin K deficiency	69
Intellectual disability, musty body odor, hypopigmented skin, eczema	Phenylketonuria (tetrahydrobiopterin [BH ₄] deficiency)	82
Bluish-black connective tissue, ear cartilage, sclerae; severe arthralgias; urine turns black on prolonged exposure to air	Alkaptonuria (homogentisate oxidase deficiency; ochronosis)	82
Infant with hypoglycemia, hepatomegaly, cardiomyopathy	Cori disease (debranching enzyme deficiency) or von Gierke disease (glucose-6-phosphatase deficiency, more severe)	85
Chronic exercise intolerance with myalgia, fatigue, painful cramps, myoglobinuria	McArdle disease (skeletal muscle glycogen phosphorylase deficiency)	85
“Cherry-red spot” on macula	Tay-Sachs (ganglioside accumulation; no hepatosplenomegaly); Niemann-Pick disease (sphingomyelin accumulation; hepatosplenomegaly); central retinal artery occlusion	86, 552
Hepatosplenomegaly, pancytopenia, osteoporosis, avascular necrosis of femoral head, bone crises	Gaucher disease (glucocerebrosidase [β-glucosidase] deficiency)	86
Achilles tendon xanthoma, corneal arcus	Familial hypercholesterolemia (↓ LDL receptor signaling)	92
Male child, recurrent infections, no mature B cells	Bruton disease (X-linked agammaglobulinemia [<i>BTK</i> gene defect])	114
Anaphylaxis following blood transfusion, atopy, airway/GI infections, autoimmune disease	Selective IgA deficiency	114
Recurrent cold (noninflamed) abscesses, eczema, ↑ serum IgE, eosinophils	Hyper-IgE syndrome (Job syndrome: neutrophil chemotaxis abnormality; <i>STAT3</i> mutation)	114
Late separation (> 30 days) of umbilical cord, no pus, recurrent skin and mucosal bacterial infections	Leukocyte adhesion deficiency (type 1; defective LFA-1 [CD18] integrin)	115
Recurrent infections and granulomas with catalase ⊕ organisms	Chronic granulomatous disease (defective NADPH oxidase)	115
Fever, vomiting, diarrhea, desquamating rash after prolonged use of nasal pack or tampon	Staphylococcal toxic shock syndrome	133
“Strawberry tongue”	Scarlet fever (sandpaper rash); Kawasaki disease (lymphadenopathy, high fever for 5 days)	134, 478
Colon cancer associated with infective endocarditis	<i>Streptococcus gallolyticus</i> (formerly <i>S bovis</i>)	135
Descending flaccid paralysis in newborn after honey ingestion	<i>Clostridium botulinum</i> infection (floppy baby syndrome)	136
Abdominal pain, diarrhea, leukocytosis, recent antibiotic use	<i>Clostridioides difficile</i> infection	136
Grayish-white pseudomembranous pharyngitis with “bull’s neck” appearance	<i>Corynebacterium diphtheriae</i> infection	137
Back pain, fever, night sweats	Pott disease (vertebral TB)	138
Acute adrenal insufficiency, fever, bilateral adrenal hemorrhage	Waterhouse-Friderichsen syndrome (meningococcemia)	140, 353

CLINICAL PRESENTATION	DIAGNOSIS/DISEASE	PAGE
Red “currant jelly” sputum in patients with alcohol overuse or diabetes	<i>Klebsiella pneumoniae</i>	143
Fever, chills, headache, myalgia following antibiotic treatment for syphilis	Jarisch-Herxheimer reaction (host response to sudden release of bacterial antigens)	144
Large rash with bull’s-eye appearance, flu-like symptoms	Erythema migrans from <i>Ixodes</i> tick bite (Lyme disease: <i>Borrelia bacteria</i>)	144
Ulcerated genital lesion	Nonpainful, indurated: chancre (1° syphilis, <i>Treponema pallidum</i>) Painful, with exudate: chancroid (<i>Haemophilus ducreyi</i>)	145, 180
Smooth, moist, painless, wartlike white lesions on genitals	Condylomata lata (2° syphilis)	145
Pupil accommodates but doesn’t react to light	Argyll Robertson pupil (3° syphilis/neurosyphilis)	145
Dog or cat bite resulting in infection (cellulitis, osteomyelitis)	<i>Pasteurella multocida</i> (cellulitis at inoculation site)	147
Atypical “walking pneumonia” with x-ray looking worse than the patient	<i>Mycoplasma pneumoniae</i> infection	148
Rash on palms and soles	Coxsackie A infection, Rocky Mountain spotted fever, 2° syphilis	148
Black eschar on face of patient with diabetic ketoacidosis and/or neutropenia	<i>Mucor</i> or <i>Rhizopus</i> fungal infection	150
Chorioretinitis, hydrocephalus, intracranial calcifications, +/- blueberry muffin rash	Congenital toxoplasmosis	153, 180
Pruritus, serpiginous rash after walking barefoot, microcytic anemia	Hookworm (<i>Ancylostoma spp</i> , <i>Necator americanus</i>)	156
Febrile child later develops red facial rash with subsequent spread to body	Erythema infectiosum/fifth disease (“slapped cheeks” appearance, caused by parvovirus B19)	161
Fever, cough, conjunctivitis, coryza, diffuse rash, Koplik spots (small, irregular red spots on buccal/lingual mucosa with blue-white centers)	Measles (rubeola) virus	167
Systolic ejection murmur (crescendo-decrescendo), “pulsus parvus et tardus,” syncope, angina, dyspnea on exertion	Aortic stenosis	296
Hyperdynamic pulses, wide pulse pressure, early diastolic murmur (decrescendo), head bobbing	Aortic regurgitation	296
Continuous “machinelike” murmur	PDA (close with indomethacin; keep open with PGE1 and PGE2 analogs)	296
Chest pain on exertion	Angina (stable: with moderate exertion; unstable: with minimal exertion or at rest)	308
Chest pain with ST depressions on ECG	Angina (⊖ troponins) or NSTEMI (⊕ troponins)	308
Chest pain, pericardial effusion/friction rub, persistent fever following MI	Postcardiac injury syndrome (autoimmune-mediated post-MI pericarditis, weeks to several months after acute episode)	314
Distant heart sounds, distended neck veins, hypotension	Beck triad of cardiac tamponade	317
Painful, raised red/purple lesions on pads of fingers/toes	Osler nodes (immune complex deposition in infective endocarditis)	318

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Painless erythematous lesions on palms and soles	Janeway lesions (septic emboli/microabscesses in infective endocarditis)	318
Splinter hemorrhages in fingernails	Infective endocarditis	318
Retinal hemorrhages with pale centers	Roth spots (infective endocarditis)	318
Telangiectasias, recurrent epistaxis, skin discoloration, arteriovenous malformations, GI bleeding, hematuria	Hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu syndrome)	320
Polyuria, polydipsia	Primary polydipsia, diabetes mellitus (type 1 or 2), diabetes insipidus (central, nephrogenic)	342, 350
No lactation postpartum, absent menstruation, cold intolerance	Sheehan syndrome (severe postpartum hemorrhage → pituitary infarction)	343
Heat intolerance, weight loss, palpitations, fine tremor, hyperreflexia	Hyperthyroidism	344
Cold intolerance, weight gain, brittle hair, depressed mood, hyporeflexia	Hypothyroidism	344
Cutaneous/dermal edema due to deposition of mucopolysaccharides in connective tissue	Myxedema (caused by hypothyroidism or hyperthyroidism [Graves disease])	344
Facial muscle spasm upon tapping	Chvostek sign (hypocalcemia)	348
Carpal spasm upon BP cuff inflation	Trousseau sign (hypocalcemia)	348
Rapid, deep, labored breathing/hyperventilation	Diabetic ketoacidosis (Kussmaul respirations)	351
Skin hyperpigmentation, orthostatic hypotension, fatigue, weakness, muscle aches, weight loss, GI disturbances	Chronic 1° adrenal insufficiency (Addison disease) → ↑ ACTH, ↑ MSH	353
Shock, altered mental status, vomiting, abdominal pain, weakness, fatigue in patient on glucocorticoid therapy	Acute adrenal insufficiency (adrenal crisis)	353
Pancreatic, pituitary, parathyroid tumors	MEN1 (autosomal dominant <i>MEN1</i> mutation)	356
Medullary thyroid carcinoma, parathyroid hyperplasia, pheochromocytoma	MEN2A (autosomal dominant <i>RET</i> mutation)	356
Medullary thyroid carcinoma, pheochromocytoma, mucosal neuromas, marfanoid habitus	MEN2B (autosomal dominant <i>RET</i> mutation)	356
Cutaneous flushing, diarrhea, bronchospasm, right-sided heart murmur	Carcinoid syndrome (urinary 5-HIAA); indicates systemic dissemination (eg, post–liver metastases)	357
Jaundice, palpable distended nontender gallbladder	Courvoisier sign (distal obstruction of common bile duct by pancreatic head malignancy)	375, 405
Hematemesis +/- abdominal/back pain	Mallory-Weiss syndrome (partial thickness esophageal lacerations in a patient with alcohol use disorder, bulimia nervosa)	384
Dysphagia (esophageal webs), glossitis, iron deficiency anemia	Plummer-Vinson syndrome (may progress to esophageal squamous cell carcinoma)	384
Enlarged, firm left supraclavicular node	Virchow node (metastasis from stomach malignancy)	386
Hematemesis, melena	Upper GI bleed (eg, peptic ulcer disease)	387
Hematochezia	Lower GI bleed (eg, colonic diverticulosis)	387
Arthralgias, cardiac and neurological symptoms, diarrhea	Whipple disease (<i>Tropheryma whipplei</i>)	388
Severe RLQ pain with palpation of LLQ	Rovsing sign (acute appendicitis)	390
Severe RLQ pain with deep palpation	McBurney sign (acute appendicitis)	390

CLINICAL PRESENTATION	DIAGNOSIS/DISEASE	PAGE
Hamartomatous GI polyps, hyperpigmented macules on mouth, lips, hands, genitalia	Peutz-Jeghers syndrome (autosomal dominant, benign polyposis can cause bowel obstruction; ↑ breast/GI cancer risk)	394
Thousands of colon polyps, osteomas/soft tissue tumors, impacted/supernumerary teeth	Gardner syndrome (subtype of familial adenomatous polyposis)	394
Severe jaundice, kernicterus (neurological changes) in neonate	Crigler-Najjar syndrome, type I (congenital unconjugated hyperbilirubinemia)	401
Golden brown rings around peripheral cornea	Wilson disease (Kayser-Fleischer rings due to copper accumulation)	402
Female, fat (obese), fertile (multiparity), forty, fair, feeds (TPN), fasting (rapid weight loss)	Cholelithiasis (gallstones)	403
Bluish line on gingiva	Burton line (lead poisoning)	425
Short stature, café-au-lait spots, thumb/radial defects, ↑ incidence of tumors/leukemia, aplastic anemia (pancytopenia)	Fanconi anemia (genetic defect in DNA crosslink repair; often progresses to AML)	427
Red/pink urine in the morning, pancytopenia, venous thrombosis	Paroxysmal nocturnal hemoglobinuria	428
Painful blue fingers/toes, hemolytic anemia	Cold autoimmune hemolytic anemia (caused by <i>Mycoplasma pneumoniae</i> , infectious mononucleosis, CLL)	429
Petechiae, mucosal bleeding, ↑ bleeding time	Platelet disorders (eg, Glanzmann thrombasthenia, Bernard Soulier, HUS, TTP, ITP, uremic platelet dysfunction)	432
Low-grade fever, night sweats, weight loss	“B” symptoms of malignancy	434
Erythematous patches → plaques → tumors	Mycosis fungoides (cutaneous T-cell lymphoma) or	435
Neonate with arm in “waiter’s tip” position following difficult birth	Erb palsy (superior trunk [C5-C6] brachial plexus injury)	452
Anterior drawer sign ⊕ (tibia glides anteriorly with respect to femur when knee is at 90° angle)	Anterior cruciate ligament (ACL) injury	454
Mosaic pattern of bone, long bone chalk-stick fractures, skull thickening, hearing loss	Osteitis deformans (Paget disease of bone, ↑ osteoblastic and osteoclastic activity)	468
Swollen, hard, painful finger joints in an elderly individual, pain worse with activity	Osteoarthritis (osteophytes on PIP [Bouchard nodes], DIP [Heberden nodes])	472
Sudden swollen/painful big toe joint, tophi	Gout/podagra (hyperuricemia)	473
Dry eyes and mouth, arthritis, parotid enlargement	Sjögren syndrome (autoimmune destruction of exocrine glands)	474
Urethritis, conjunctivitis, arthritis	Reactive arthritis associated with HLA-B27	475
“Butterfly” facial rash, arthritis, cytopenia, and fever in a female of reproductive age	Systemic lupus erythematosus	476
Cervical lymphadenopathy, desquamating rash, coronary aneurysms, red conjunctivae and tongue, hand-foot changes (edema, erythema), fever ≥ 5 days	Kawasaki disease (mucocutaneous lymph node syndrome, treat with IVIG and aspirin)	478
Palpable purpura on buttocks/legs, joint pain, abdominal pain, hematuria in a child	IgA vasculitis (Henoch-Schönlein purpura, affects skin and kidneys)	479

CLINICAL PRESENTATION	DIAGNOSIS/DISEASE	PAGE
Painful fingers/toes changing color from white to blue to red with cold or stress	Raynaud phenomenon (vasospasm in extremities)	480
Dark purple skin/mouth nodules in a patient with AIDS	Kaposi sarcoma, associated with HHV-8	486
Pruritic, purple, polygonal planar papules and plaques (6 Ps)	Lichen planus	492
Dorsiflexion of large toe with fanning of other toes upon plantar scrape	Babinski sign (UMN lesion)	523, 543
Intention tremor, limb ataxia, loss of balance (fall toward injured side)	Cerebellar lesion (hemispheric affects voluntary movement of extremities; vermis affects axial and proximal movement)	524
Hyperphagia, hypersexuality, hyperorality	Klüver-Bucy syndrome (bilateral amygdala lesion; HSV-1 encephalitis)	524
Resting tremor, athetosis, chorea	Basal ganglia lesion (eg, Huntington disease, Parkinson disease)	524
Dysphagia, hoarseness, ↓ gag reflex, nystagmus, ipsilateral Horner syndrome	Lateral medullary (Wallenberg) syndrome (posterior inferior cerebellar artery lesion)	527
Lucid interval after traumatic brain injury	Epidural hematoma (middle meningeal artery rupture; branch of maxillary artery; lentiform collection not crossing suture lines)	528
“Worst headache of my life”	Subarachnoid hemorrhage	528
Resting tremor, rigidity, akinesia (shuffling gait, micrographia), postural instability	Parkinson disease (loss of dopaminergic neurons in substantia nigra pars compacta)	534
Chorea, dementia, caudate degeneration	Huntington disease (loss of GABAergic neurons in striatum; autosomal dominant CAG repeat expansion)	534
Urinary incontinence, gait apraxia, cognitive dysfunction	Normal pressure hydrocephalus (ventricle expansion → distortion of corona radiata fibers)	536
Relapsing and remitting nystagmus, intention tremor, optic neuritis, scanning speech, bilateral internuclear ophthalmoplegia	Multiple sclerosis	537
Progressive, ascending symmetric limb weakness and hyporeflexia after GI/upper respiratory infection	Guillain-Barré syndrome (acute inflammatory demyelinating polyneuropathy)	538
Café-au-lait spots, Lisch nodules (iris hamartomas), cutaneous neurofibromas, pheochromocytomas, optic gliomas	Neurofibromatosis type I	539
Bilateral vestibular schwannomas	Neurofibromatosis type II	539
Vascular birthmark (port-wine stain) on face	Nevus flammeus (benign, associated with Sturge-Weber syndrome)	539
Bilateral renal cell carcinomas, hemangioblastomas, angiomas, pheochromocytomas	von Hippel-Lindau disease (VHL deletion on chromosome 3p)	539
Hyperreflexia, hypertonia, ⊕ Babinski sign	UMN damage	543
Hyporeflexia, hypotonia, atrophy, fasciculations	LMN damage	543
Staggering gait, frequent falls, nystagmus, hammer toes, diabetes mellitus, hypertrophic cardiomyopathy	Friedreich ataxia	545
Unilateral facial drooping involving forehead	LMN facial nerve (CN VII) palsy; UMN lesions spare forehead due to bilateral innervation	546

CLINICAL PRESENTATION	DIAGNOSIS/DISEASE	PAGE
Episodic vertigo, tinnitus, sensorineural hearing loss	Ménière disease (↑ endolymph in inner ear)	548
Ptosis, miosis, anhidrosis	Horner syndrome (sympathetic chain lesion)	555
Conjugate horizontal gaze palsy, horizontal diplopia	Internuclear ophthalmoplegia (MLF lesion; unilateral or bilateral)	558
Acute onset, reversible “waxing and waning” level of consciousness (acute onset), ↓ attention span, ↓ arousal	Delirium (usually 2° to other cause)	575
Polyuria, renal tubular acidosis type II, growth retardation, electrolyte imbalances, hypophosphatemic rickets	Fanconi syndrome (generalized reabsorption defect of proximal convoluted tubule)	604, 611
Periorbital and/or peripheral edema, proteinuria (> 3.5 g/day; frothy urine), hypoalbuminemia, hypercholesterolemia	Nephrotic syndrome (podocyte damage)	613
Hereditary nephritis, sensorineural hearing loss, retinopathy, anterior lenticonus	Alport syndrome (type IV collagen mutation)	615
Wilms tumor, macroglossia, organomegaly, hemihyperplasia, omphalocele	Beckwith-Wiedemann syndrome (WT2 mutation)	624
Streak ovaries, congenital heart disease, horseshoe kidney, cystic hygroma, short stature, webbed neck, lymphedema	Turner syndrome (45,XO)	655
Ovarian fibroma, ascites, pleural effusion	Meigs syndrome	665
Red, itchy, swollen rash of nipple/areola	Paget disease of the breast (sign of underlying neoplasm)	668
Fibrous plaques in tunica albuginea of penis with abnormal curvature	Peyronie disease (associated with repeated minor trauma during intercourse)	669
Hypoxemia, polycythemia, hypercapnia	Chronic bronchitis (hypertrophy and hyperplasia of mucus-secreting glands)	692
Pink complexion, dyspnea, hyperventilation	Emphysema (centriacinar [tobacco smoking] or panacinar [α_1 -antitrypsin deficiency])	692
Bilateral hilar adenopathy, uveitis, arthropathy, skin changes	Sarcoidosis (immune-mediated noncaseating granulomas)	695

► CLASSIC LABS/FINDINGS

LAB/DIAGNOSTIC FINDING	DIAGNOSIS/DISEASE	PAGE
Colonies of <i>Pseudomonas</i> in lungs, ↑ Cl^- on sweat test, ↑ immunoreactive trypsinogen	Cystic fibrosis (autosomal recessive mutation in <i>CFTR</i> gene → fat-soluble vitamin deficiency and mucous plugs)	58
↓ AFP on second trimester screening	Down syndrome, Edwards syndrome	61
↑ β -hCG, ↓ PAPP-A on first trimester screening	Down syndrome	61
Homocysteine, methylmalonic acid, folate	Vitamin B ₁₂ deficiency	67
Anti-histone antibodies	Drug-induced lupus	113
↓ T cells, ↓ PTH, ↓ Ca^{2+} , absent thymic shadow on CXR	Thymic aplasia (22q11 microdeletion: DiGeorge syndrome, velocardiofacial syndrome)	114
Recurrent infections, eczema, thrombocytopenia, ↑ IgE and IgA, ↓/normal IgG and IgM	Wiskott-Aldrich syndrome (WAS gene mutation)	115